

Bayesian Modeling of Customer Churn Probability with Hierarchical Extensions

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1 Introduction

Customer churn is a central metric for subscription-based services, as it directly affects long-term revenue and operational planning. Understanding not only the overall likelihood of churn but also how this probability differs across customer subgroups is essential for targeted retention strategies. Bayesian modeling provides a natural and coherent framework for estimating churn probabilities, quantifying uncertainty, and incorporating hierarchical structure to borrow strength across related groups.

This project develops a sequence of Bayesian models based on the **Telco Customer Churn** dataset. We address three main research questions:

- What is the company-wide churn probability?
- Do churn probabilities differ substantially by contract type (Month-to-month, One year, Two year)?
- Does a hierarchical model better capture subgroup variation compared to a single-parameter model?

We begin with a one-parameter Beta-Binomial model, extend to a hierarchical Beta-Binomial structure, and evaluate model fit using posterior predictive checks.

2 Data Description

The Telco dataset contains information on 7043 telecom customers, including demographic variables, subscription type, and churn outcome (“Yes” or “No”). For modeling purposes:

- **Churn** is encoded as a binary variable (1 = churned, 0 = retained).

- **Contract type**, a key grouping variable with strong heterogeneity, includes three categories Month-to-month, One year, and Two year.
- Overall churn rate in the sample is approximately 26.5%.

A simple exploratory analysis reveals substantial differences across groups. Month-to-month customers churn at 43%, while One-year (11%) and Two-year (3%) customers churn much less frequently. These patterns strongly motivate a hierarchical modeling approach. Visual summaries are provided in Figure 1 and Figure 2.

3 Method

3.1 One-Parameter Beta-Binomial Model

In the research, we set

$$Y_i|\theta \sim \text{Bernoulli}(\theta),$$

where $Y_i = 1$ if customer i churned and 0 otherwise, and θ is the (unknown) population-level churn probability we wish to estimate. We are therefore interested in the posterior distribution of θ given the observed churn data.

Let n denote the sample size and Y denote the total number of churned customers in the sample dataset. Then Y follows Binomial distribution given n and θ , i.e.

$$Y|\theta \sim \text{Binomial}(n, \theta).$$

We choose a conjugate Beta prior:

$$\theta \sim \text{Beta}(a, b).$$

Then using conjugacy and observing n customers with Y churns, posterior is

$$\theta|Y \sim \text{Beta}(a + Y, b + n - Y).$$

Using a uniform prior $\theta \sim \text{Beta}(1, 1)$, posterior is $\theta|Y \sim \text{Beta}(1 + Y, 1 + n - Y)$.

Monte Carlo approximation is implemented via simulation from the posterior Beta distribution, enabling estimation of means, credible intervals, and posterior predictive distributions.

3.2 Hierarchical Beta-Binomial Model

To model variation across contract types, define groups $j = 1, 2, 3$ with churn counts Y_j and group sizes n_j . We introduce group-specific churn probabilities:

$$Y_j|\theta_j \sim \text{Binomial}(n_j, \theta_j), \quad j = 1, 2, 3,$$

where $j = 1$ represents Month-to-month, $j = 2$ represents One year, and $j = 3$ represents Two year. The group-level parameters follow a common Beta prior:

$$\theta_j \sim \text{Beta}(\alpha, \beta), \quad j = 1, 2, 3.$$

with hyperpriors:

$$\alpha, \beta \sim \text{Exponential}(1).$$

This structure allows partial pooling: groups with smaller sample sizes shrink toward the common mean, while large groups express stronger individual signals. Posterior inference is carried out using Hamiltonian Monte Carlo (Stan), with θ_j , α , β jointly estimated.

3.3 Posterior Predictive Checks

A posterior predictive check (PPC) evaluates whether data simulated from the fitted model look like the observed data. The research simulates replicated churn counts $\hat{Y}$ from the posterior predictive distribution:

$$\hat{Y}|Y \sim \text{Binomial}(n, \theta^*)$$

for the one-parameter model, and:

$$\hat{Y}_j|Y_j \sim \text{Binomial}(n_j, \theta_j^*)$$

for hierarchical modeling.

4 Results

4.1 One-Parameter Model

Across all 7,043 customers, a total of 1,869 churn events were observed, yielding an empirical churn rate of 26.5%.

Under the one-parameter Beta-Binomial model with a uniform prior $\text{Beta}(1, 1)$, the posterior distribution becomes

$$\theta|Y \sim \text{Beta}(1870, 5175),$$

for which the posterior mean is

$$E[\theta|Y] \approx 0.265,$$

and the 95% confidence interval is approximately $[0.255, 0.276]$. These values closely match the empirical churn rate, which is expected given the large sample size.

Posterior Monte Carlo draws (Figure 3) form a tightly concentrated distribution, indicating that the overall churn probability is estimated with high precision. The posterior predictive distribution for the total churn count (Figure 4) shows that the observed churn total falls near the center of the predictive histogram, with a two-sided posterior predictive p -value of approximately 0.99. This indicates an excellent fit at the aggregate level, suggesting that a single-binomial model is sufficient for describing overall churn behavior.

However, visual inspection of churn rates by subgroup (Figure 2) reveals striking variation across contract types, which a one-parameter model is unable to capture. This motivates the hierarchical modeling approach presented next.

4.2 Hierarchical Model

Contract-level summaries show large differences in churn behavior:

- Month-to-month customers (1,655 churns out of 3,875): $\theta_1 \approx 0.427$,
- One-year customers (166 churns out of 1,473): $\theta_2 \approx 0.113$,
- Two-year customers (48 churns out of 1,695): $\theta_3 \approx 0.029$.

1 These differences are substantial enough that a single churn probability θ is not appropriate.

The contract-level posterior densities (Figure 5) show almost no overlap between groups, indicating strong evidence for heterogeneity. Credible intervals for each group are shown in Figure 6.

The estimated hyperparameters,

$$\alpha \approx 0.63, \beta \approx 1.80,$$

indicate a wide underlying Beta distribution, reflecting large intrinsic variability among contract types. The hierarchical posterior predictive distributions (Figure 7) align closely with the observed churn counts for each contract category. In contrast, the one-parameter model significantly underestimates churn for Month-to-month customers and overestimates it for One-year and Two-year customers. This discrepancy is clearly illustrated in the model-comparison plot (Figure 8), where the one-parameter model pulls all predicted churn counts toward a global mean and fails to reproduce observed extremes.

Additionally, posterior predictive p -values for each contract type are near 1.0, which indicates that, unlike the single-parameter model, the hierarchical model produces realistic contract-level predictions that closely match empirical churn patterns.

5 Conclusion

This analysis demonstrates the value of Bayesian modeling for understanding churn probability in subscription-based services. The one-parameter Beta-Binomial model provides a precise estimate of the overall churn probability (approximately 26.5%), and posterior predictive checks show that it fits the aggregate churn count extremely well.

However, this model fails to account for the strong subgroup heterogeneity observed across contract types. The hierarchical Beta-Binomial model successfully captures this variation, producing posterior estimates of churn probability that differ by more than an order of magnitude—from roughly 43% for Month-to-month customers to just 3% for Two-year customers. Posterior predictive checks

further confirm that the hierarchical model accurately reproduces contract-level churn patterns, while the one-parameter model substantially mispredicts them.

These findings highlight the necessity of subgroup modeling in churn analysis and the advantage of Bayesian hierarchical approaches for structured data.

Overall, Bayesian hierarchical modeling offers a powerful framework for churn analysis, supporting more realistic risk estimation and more effective segmentation strategies.

6 Limitations

- Only contract type was modeled hierarchically; other important grouping variables (e.g., InternetService, PaymentMethod) were not incorporated.
- Customer-level covariates were omitted; the model does not capture individual-level risk factors.
- The Beta-Binomial assumption may oversimplify churn behavior by ignoring potential temporal patterns.

7 Future Work

- Extend the model to multiple hierarchical factors or cross-classified structures.
- Incorporate customer-level predictors using Bayesian regression.
- Explore more flexible priors (e.g., hierarchical Dirichlet processes) to discover latent customer segments.
- Integrate temporal or tenure-based components into the model. Churn is inherently time-dependent, and survival or hazard-based Bayesian models could capture how churn probability evolves over a customer's lifecycle.

Appendix: Figures

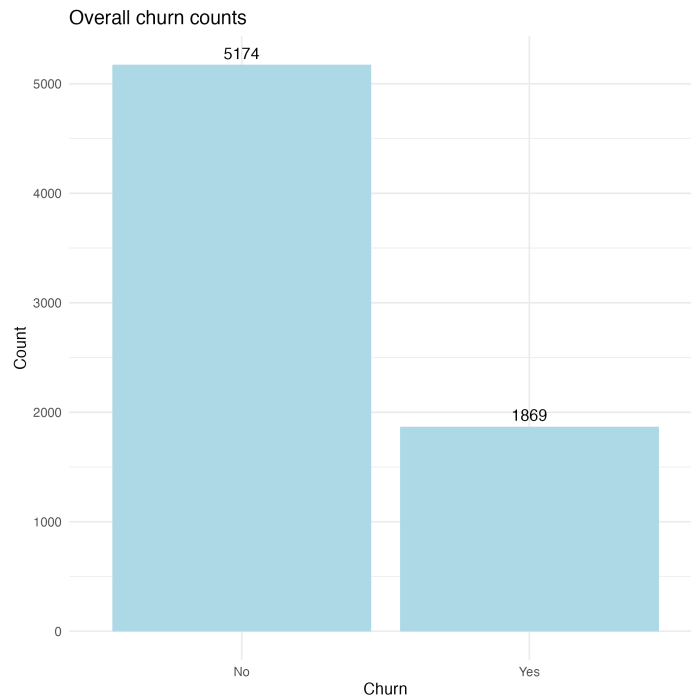


Figure 1: Overall churn counts.

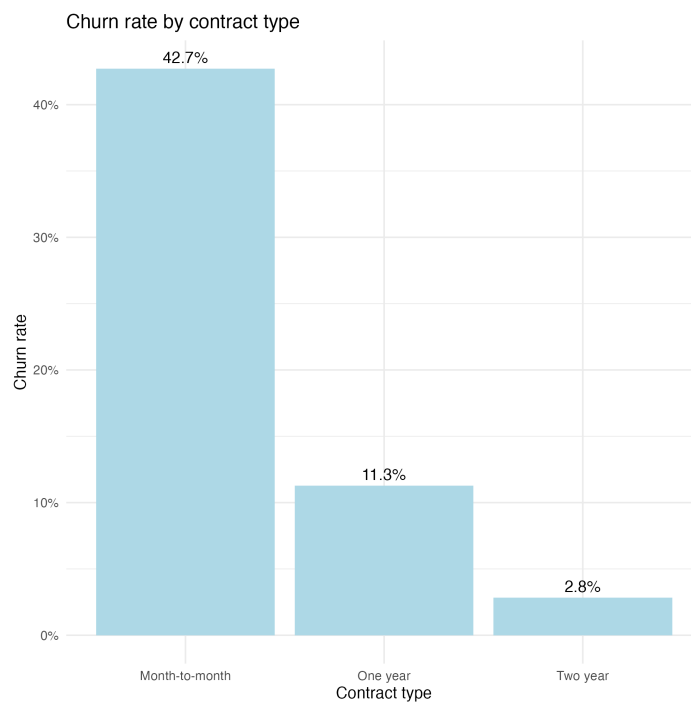


Figure 2: Churn rate by contract type.

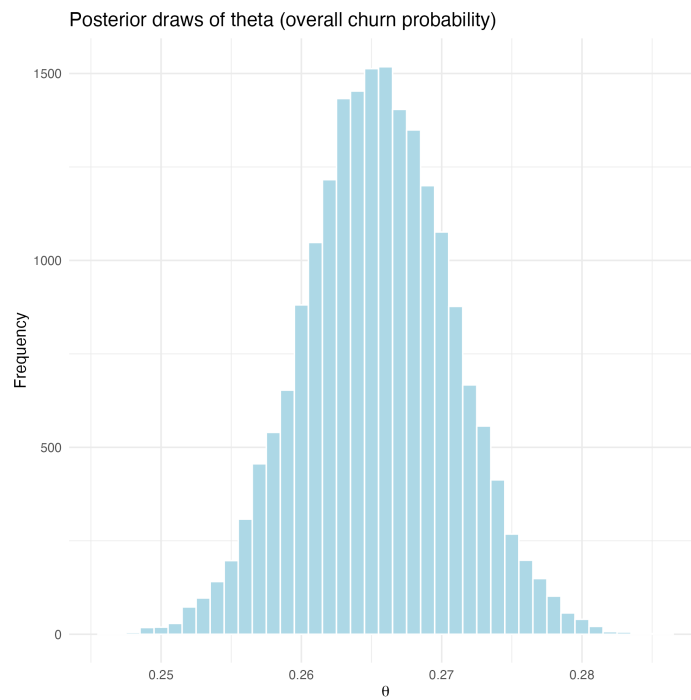


Figure 3: Monte Carlo approximation from posterior Beta.

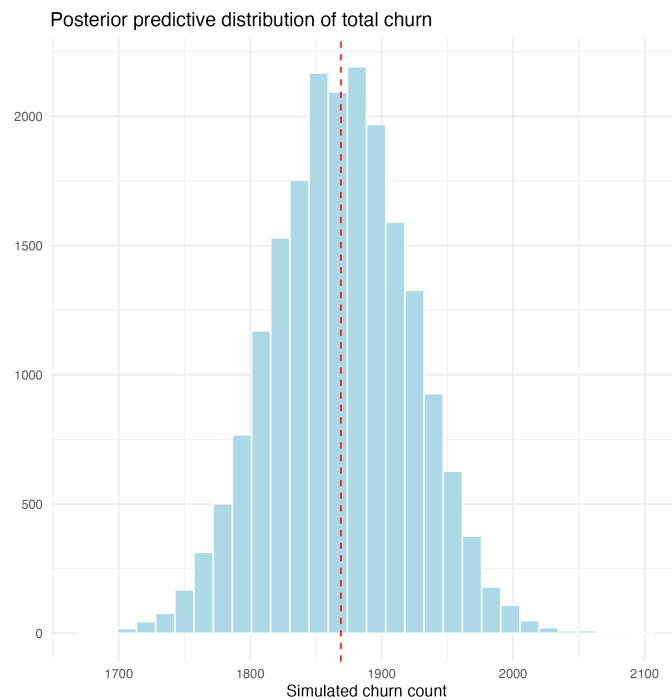


Figure 4: Posterior predictive distribution for total churn count.

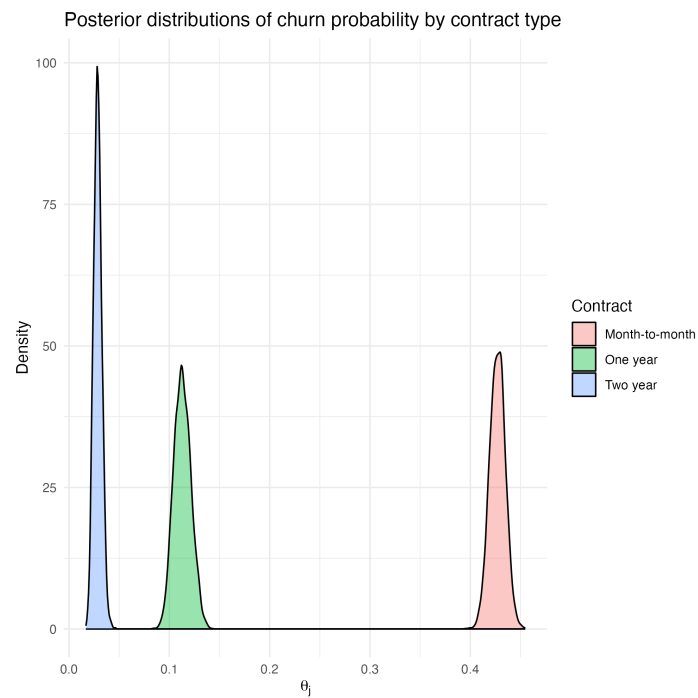


Figure 5: Posterior density of churn probability by contract type.

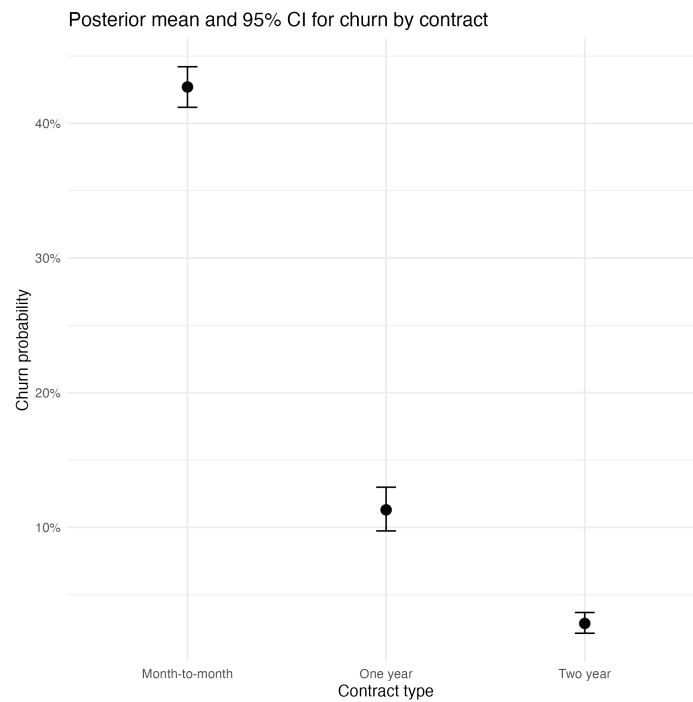


Figure 6: Posterior means and 95% credible intervals for group-level churn probabilities.

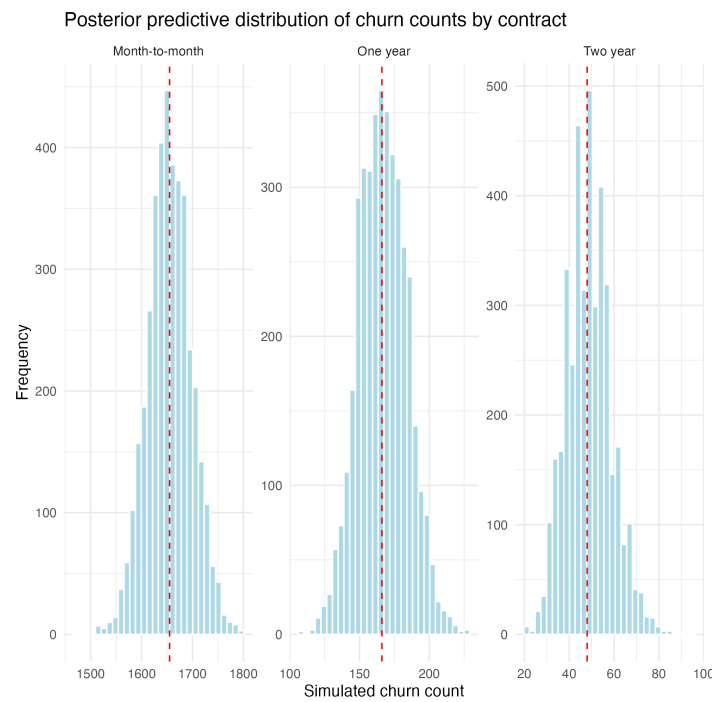


Figure 7: Posterior predictive check for churn counts by contract type.

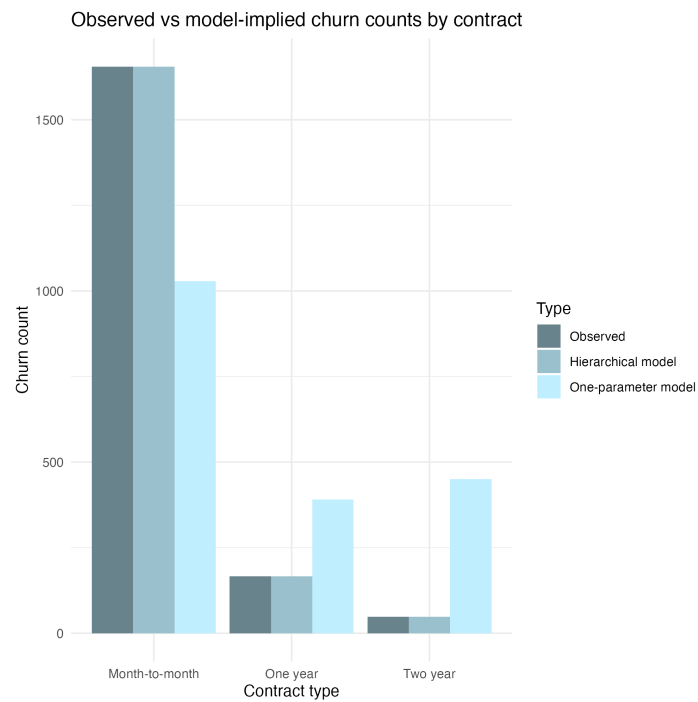


Figure 8: Compare one-parameter vs hierarchical at group level.

Appendix: Code

```
In [31]: library(tidyverse)
library(rstan)
library(bayesplot)

options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
theme_set(theme_minimal())
```

Load Data and Preprocessing

```
In [32]: telco_raw <- read_csv("Telco-Customer-Churn.csv",
                               show_col_types = FALSE)

telco <- telco_raw %>%
  mutate(
    Churn = if_else(Churn == "Yes", 1L, 0L),
    Contract = factor(Contract),
    InternetService = factor(InternetService),
    PaymentMethod = factor(`PaymentMethod`)
  )

glimpse(telco)
unique(telco$Contract)
```

```
Rows: 7,043
Columns: 21
$ customerID    <chr> "7590-VHVEG", "5575-GNVDE", "3668-QPYBK", "7795-CF0CW...
$ gender        <chr> "Female", "Male", "Male", "Male", "Female", "Female",...
$ SeniorCitizen <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ Partner       <chr> "Yes", "No", "No", "No", "No", "No", "No", "No", "No", "Yes...
$ Dependents    <chr> "No", "No", "No", "No", "No", "No", "Yes", "No", "No"...
$ tenure        <dbl> 1, 34, 2, 45, 2, 8, 22, 10, 28, 62, 13, 16, 58, 49, 2...
$ PhoneService  <chr> "No", "Yes", "Yes", "No", "Yes", "Yes", "Yes", "No", ...
$ MultipleLines <chr> "No phone service", "No", "No", "No phone service", "...
$ InternetService <fct> DSL, DSL, DSL, DSL, Fiber optic, Fiber optic, Fiber o...
$ OnlineSecurity <chr> "No", "Yes", "Yes", "Yes", "No", "No", "No", "Yes", "...
$ OnlineBackup  <chr> "Yes", "No", "Yes", "No", "No", "No", "Yes", "No", "N...
$ DeviceProtection <chr> "No", "Yes", "No", "Yes", "No", "Yes", "No", "No", "Y...
$ TechSupport   <chr> "No", "No", "No", "Yes", "No", "No", "No", "No", "Yes...
$ StreamingTV   <chr> "No", "No", "No", "No", "No", "Yes", "Yes", "No", "Ye...
$ StreamingMovies <chr> "No", "No", "No", "No", "No", "Yes", "No", "No", "Yes...
$ Contract      <fct> Month-to-month, One year, Month-to-month, One year, M...
$ PaperlessBilling <chr> "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes", "No", ...
$ PaymentMethod <fct> Electronic check, Mailed check, Mailed check, Bank tr...
$ MonthlyCharges <dbl> 29.85, 56.95, 53.85, 42.30, 70.70, 99.65, 89.10, 29.7...
$ TotalCharges  <dbl> 29.85, 1889.50, 108.15, 1840.75, 151.65, 820.50, 1949...
$ Churn         <int> 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0,...
```

Month-to-month · One year · Two year

► Levels:

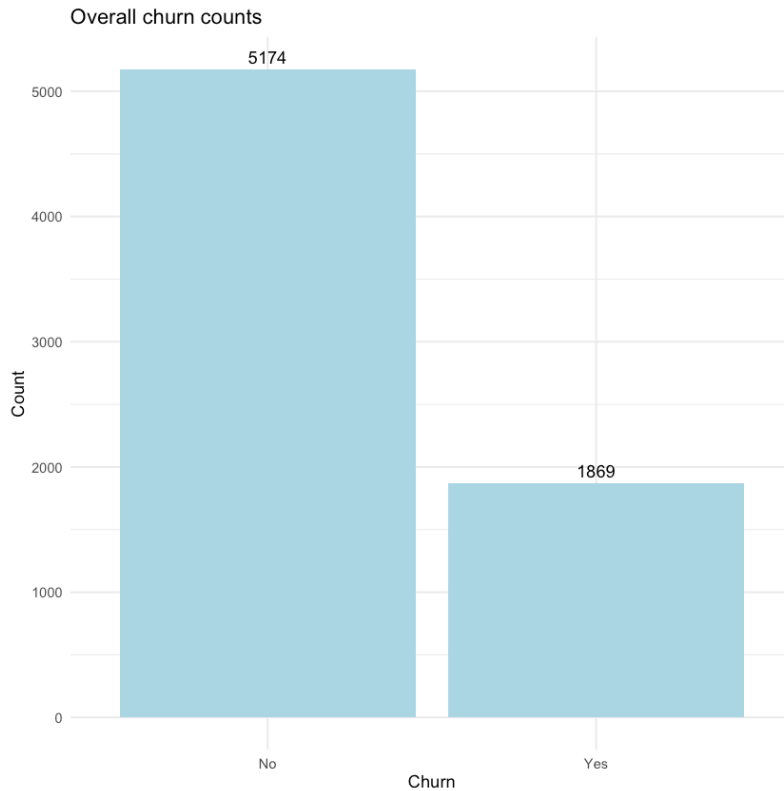
Exploratory Data Visualization

```
In [33]: # Overall churn vs non-churn counts
plot_churn_bar <- telco %>%
```

```
count(Churn) %>%
mutate(Churn = factor(Churn, levels = c(0, 1),
                      labels = c("No", "Yes"))) %>%
ggplot(aes(x = Churn, y = n)) +
geom_col(fill = "lightblue") +
geom_text(aes(label = n), vjust = -0.5) +
labs(title = "Overall churn counts",
      x = "Churn", y = "Count")

plot_churn_bar
ggsave("churn_bar.png", plot_churn_bar, dpi = 300)
```

Saving 6.67 x 6.67 in image

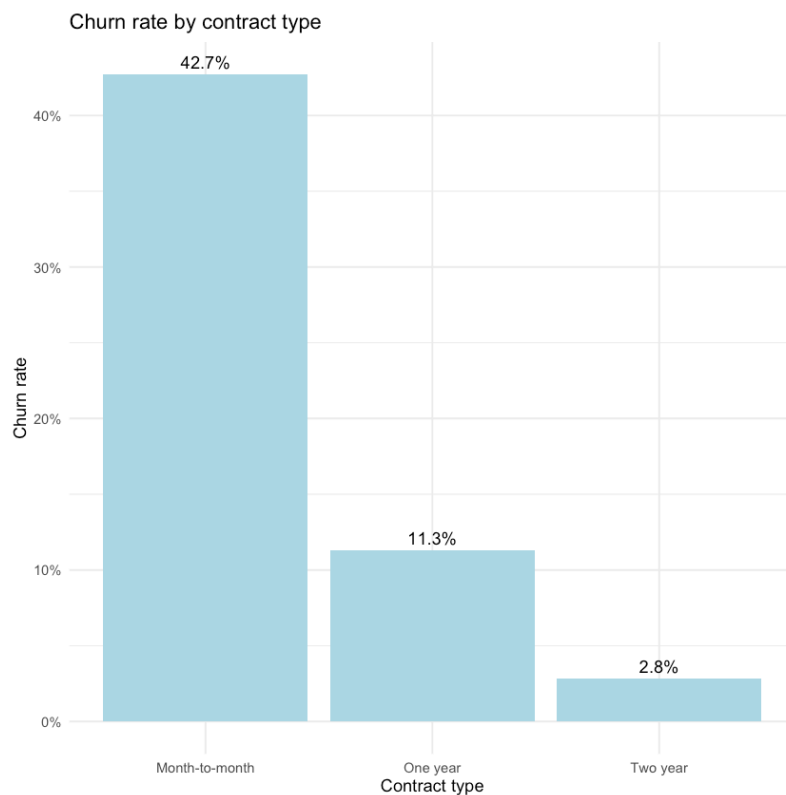


```
In [34]: # Churn rate by Contract type
churn_by_contract <- telco %>%
  group_by(Contract) %>%
  summarise(
    churn_rate = mean(Churn),
    n = n(),
    .groups = "drop"
  )

plot_contract_rate <- churn_by_contract %>%
ggplot(aes(x = Contract, y = churn_rate)) +
geom_col(fill = "lightblue") +
geom_text(aes(label = scales::percent(churn_rate, accuracy = 0.1)),
          vjust = -0.5) +
labs(title = "Churn rate by contract type",
      x = "Contract type", y = "Churn rate") +
scale_y_continuous(labels = scales::percent_format())

plot_contract_rate
ggsave("contract_rate.png", plot_contract_rate, dpi = 300)
```

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One-parameter Bayesian model: Beta-Binomial

```
In [35]: n <- nrow(telco)
y <- sum(telco$Churn)

cat("Sample size:", n, "\nNumber of churns:", y, "\nOverall churn rate:", y / n)

Sample size: 7043
Number of churns: 1869
Overall churn rate: 0.2653699
```

```
In [36]: set.seed(321)
a <- 1; b <- 1

post_a <- a + y
post_b <- b + (n - y)

post_mean <- post_a / (post_a + post_b)
ci <- qbeta(c(0.025, 0.975), post_a, post_b)

cat("Posterior mean:", post_mean)
cat("\n95% Confidence Interval: [", ci[1], ", ", ci[2], "]")

Posterior mean: 0.2654365
95% Confidence Interval: [ 0.2551897 , 0.2758094 ]
```

```
In [37]: # Monte Carlo approximation from posterior Beta
set.seed(321)
n_sims <- 20000
theta_samples <- rbeta(n_sims, post_a, post_b)
theta_df <- data.frame(theta = theta_samples)

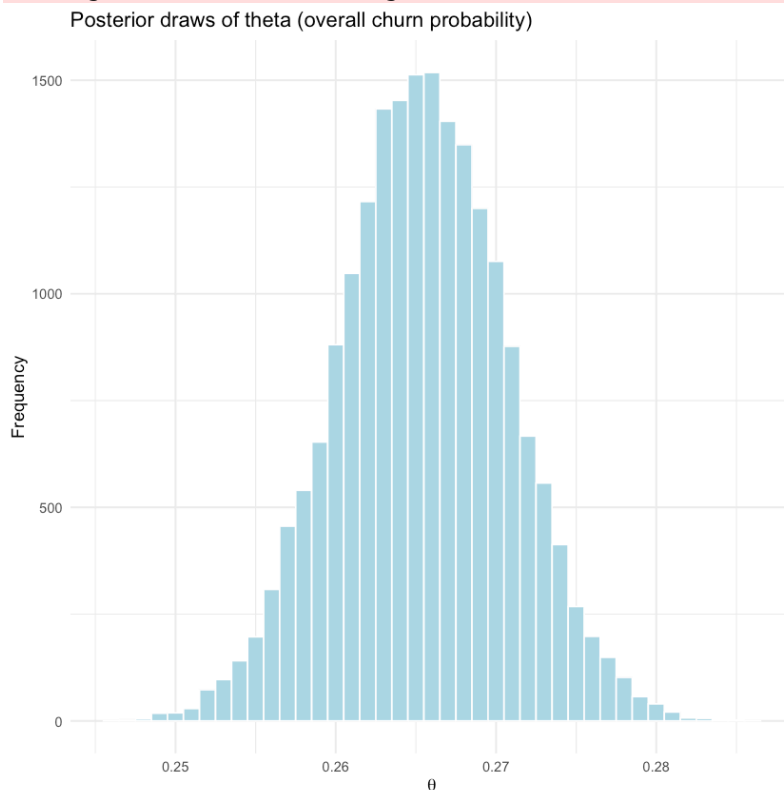
ggplot(theta_df, aes(x = theta)) +
  geom_histogram(binwidth = 0.001, fill = "lightblue", color = "white") +
  labs(
    title = "Posterior draws of theta (overall churn probability)",
    x = expression(theta),
```

```

  y = "Frequency"
) +
  theme_minimal()
ggsave("mc_posterior.png", dpi = 300)

```

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```

In [38]: # Posterior predictive distribution for churn count (overall)
set.seed(321)
y_rep_total <- rbinom(n_sims, size = n, prob = theta_samples)

ppc_total_hist <- qplot(y_rep_total,
  bins = 30,
  xlab = "Simulated churn count",
  main = "Posterior predictive distribution of total churn",
  fill = I("lightblue"), color = I("white")) +
  geom_vline(xintercept = y, color = "red", linetype = "dashed") +
  theme_minimal()

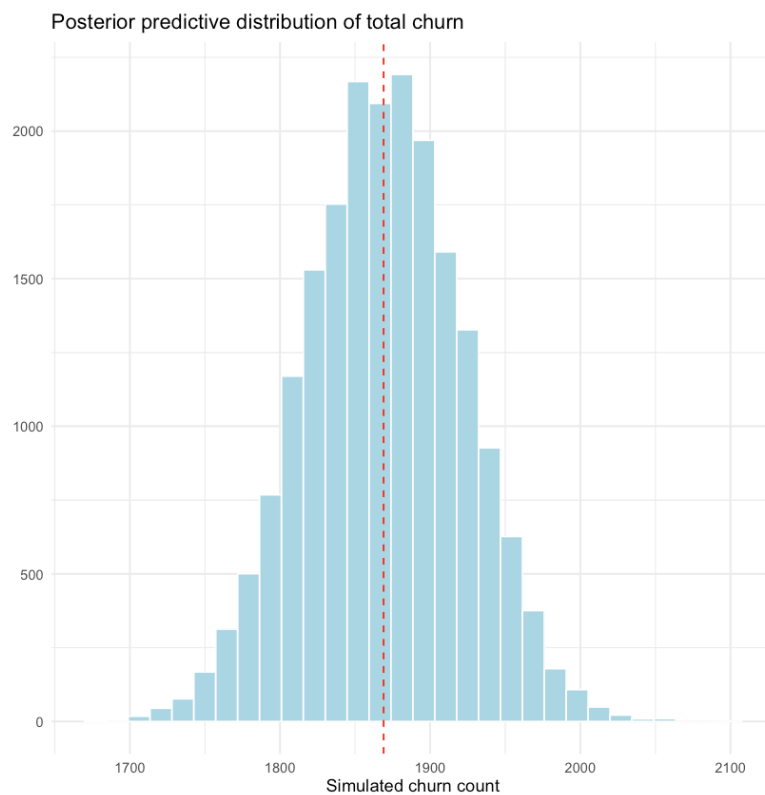
ppc_total_hist

pp_value_total <- 2 * min(
  mean(y_rep_total <= y),
  mean(y_rep_total >= y)
)
cat("Posterior predictive p-value:", pp_value_total)
ggsave("ppc_total.png", ppc_total_hist, dpi = 300)

```

Posterior predictive p-value: 1.0005

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Hierarchical Beta-Binomial model by Contract

```
In [39]: group_data <- telco %>%
  group_by(Contract) %>%
  summarise(
    n = n(),
    y = sum(Churn),
    .groups = "drop"
  )
```

group_data

```
J <- nrow(group_data)
n_j <- group_data$n
y_j <- group_data$y

stan_data_hier <- list(
  J = J,
  n = n_j,
  y = y_j
)
```

A tibble: 3 × 3

Contract	n	y
<fct>	<int>	<int>
Month-to-month	3875	1655
One year	1473	166
Two year	1695	48

```
In [40]: set.seed(321)

fit_hier <- stan(
  file = "hierarchical_model.stan",
```

```

data = stan_data_hier,
iter = 2000,
warmup = 1000,
chains = 4,
seed = 321
)
fit_hier

```

Inference for Stan model: anon_model.

4 chains, each with iter=2000; warmup=1000; thin=1;

post-warmup draws per chain=1000, total post-warmup draws=4000.

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%
alpha	0.63	0.01	0.35	0.15	0.38	0.57	0.81	1.49
beta	1.80	0.02	1.11	0.33	0.98	1.58	2.36	4.60
theta[1]	0.43	0.00	0.01	0.41	0.42	0.43	0.43	0.44
theta[2]	0.11	0.00	0.01	0.10	0.11	0.11	0.12	0.13
theta[3]	0.03	0.00	0.00	0.02	0.03	0.03	0.03	0.04
y_rep[1]	1655.03	0.67	43.08	1570.00	1627.00	1654.00	1684.00	1742.00
y_rep[2]	166.73	0.25	17.33	134.00	155.00	166.00	178.00	201.00
y_rep[3]	48.81	0.14	9.92	31.00	42.00	48.50	55.00	70.00
lp__	-3392.08	0.04	1.59	-3396.04	-3392.94	-3391.78	-3390.91	-3389.96

	n_eff	Rhat
alpha	4179	1
beta	4293	1
theta[1]	4924	1
theta[2]	5022	1
theta[3]	4382	1
y_rep[1]	4132	1
y_rep[2]	4624	1
y_rep[3]	4728	1
lp__	1683	1

Samples were drawn using NUTS(diag_e) at Thu Dec 11 17:23:24 2025.

For each parameter, n_eff is a crude measure of effective sample size, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat=1).

Hierarchical model: visualizing posterior θ_j

```

In [41]: set.seed(321)
posterior_draws <- rstan::extract(fit_hier)

theta_draws <- posterior_draws$theta
alpha_draws <- posterior_draws$alpha
beta_draws <- posterior_draws$beta

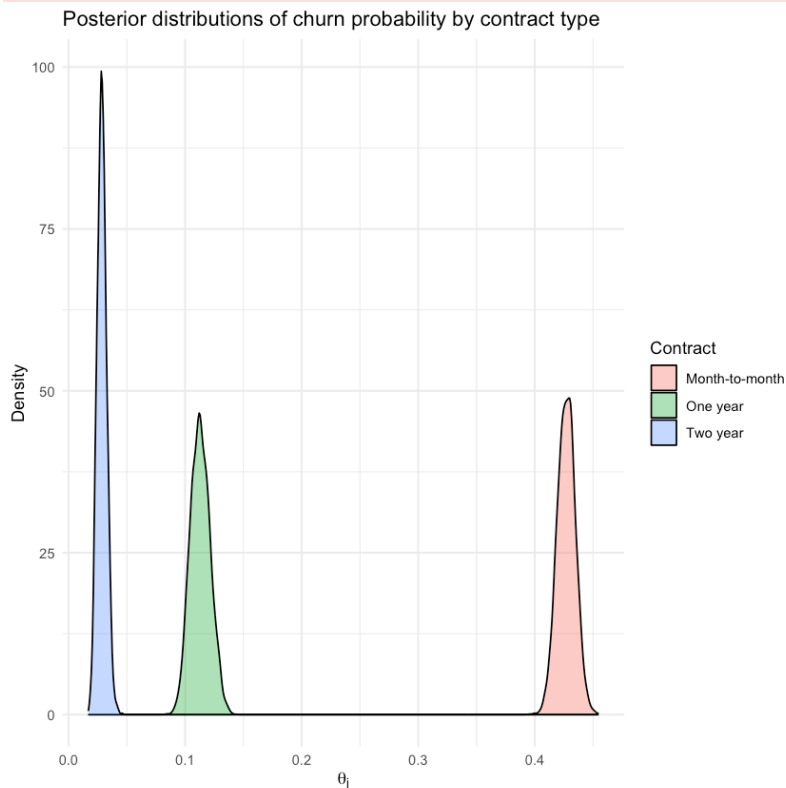
theta_df <- as_tibble(theta_draws) %>%
  setNames(as.character(group_data$Contract)) %>%
  pivot_longer(cols = everything(),
               names_to = "Contract",
               values_to = "theta")

# Density plots for each contract type
plot_theta_contract_density <- theta_df %>%
  ggplot(aes(x = theta, fill = Contract)) +
  geom_density(alpha = 0.4) +
  labs(title = "Posterior distributions of churn probability by contract type",
       x = expression(theta[j]),
       y = "Density")

plot_theta_contract_density
ggsave("theta_density.png", plot_theta_contract_density, dpi = 300)

```

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```
In [42]: # Posterior means & 95% intervals for theta_j
theta_summary <- theta_df %>%
  group_by(Contract) %>%
  summarise(
    mean = mean(theta),
    lower = quantile(theta, 0.025),
    upper = quantile(theta, 0.975),
    .groups = "drop"
  )

theta_summary

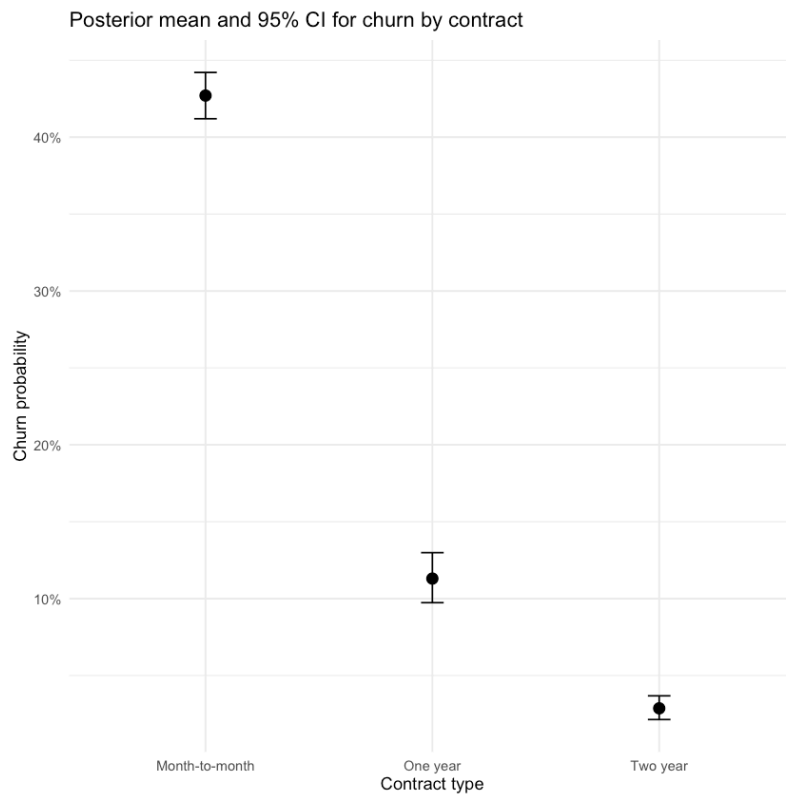
plot_theta_contract_interval <- theta_summary %>%
  ggplot(aes(x = Contract, y = mean)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = lower, ymax = upper),
    width = 0.1) +
  labs(title = "Posterior mean and 95% CI for churn by contract",
    x = "Contract type",
    y = "Churn probability") +
  scale_y_continuous(labels = scales::percent_format())

plot_theta_contract_interval
ggsave("theta_interval.png", plot_theta_contract_interval, dpi = 300)
```

A tibble: 3 x 4

Contract	mean	lower	upper
<chr>	<dbl>	<dbl>	<dbl>
Month-to-month	0.42704493	0.41196484	0.44209521
One year	0.11300355	0.09735568	0.12982998
Two year	0.02864402	0.02136632	0.03678006

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Group-level posterior predictive check

```
In [43]: yrep_mat <- posterior_draws$y_rep

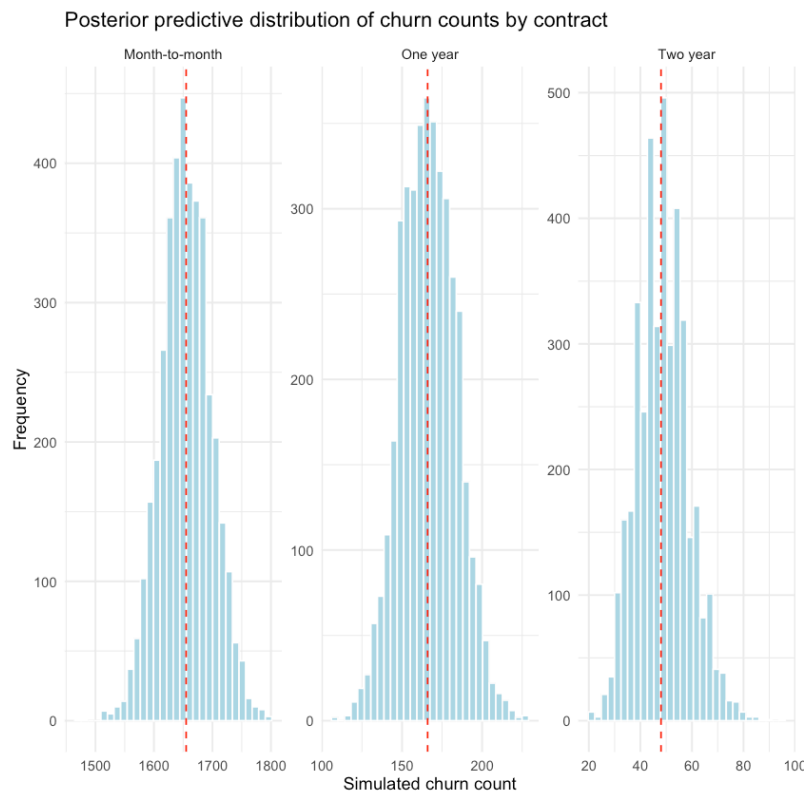
yrep_df <- as_tibble(yrep_mat) %>%
  setNames(as.character(group_data$Contract)) %>%
  pivot_longer(cols = everything(),
               names_to = "Contract",
               values_to = "y_rep")

obs_df <- group_data %>%
  select(Contract, y) %>%
  mutate(Contract = as.character(Contract))

ppc_contract <- yrep_df %>%
  ggplot(aes(x = y_rep)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  facet_wrap(~ Contract, scales = "free") +
  geom_vline(data = obs_df,
            aes(xintercept = y),
            color = "red", linetype = "dashed") +
  labs(title = "Posterior predictive distribution of churn counts by contract",
       x = "Simulated churn count",
       y = "Frequency")

ppc_contract
ggsave("ppc_contract.png", ppc_contract, dpi = 300)
```

Saving 6.67 x 6.67 in image



```
In [44]: # Compute simple PPC p-values per group
pp_contract_p <- yrep_df %>%
  group_by(Contract) %>%
  summarise(
    p_lower = mean(y_rep <= obs_df$y[match(Contract, obs_df$Contract)]),
    p_upper = mean(y_rep >= obs_df$y[match(Contract, obs_df$Contract)]),
    .groups = "drop"
  ) %>%
  mutate(p_two_sided = 2 * pmin(p_lower, p_upper))

pp_contract_p
```

A tibble: 3 × 4

Contract	p_lower	p_upper	p_two_sided
<chr>	<dbl>	<dbl>	<dbl>
Month-to-month	0.51450	0.4960	0.9920
One year	0.50475	0.5175	1.0095
Two year	0.50000	0.5370	1.0000

Compare one-parameter vs hierarchical at group level

```
In [45]: # One-parameter posterior mean theta (from Model 1)
theta_overall_hat <- post_mean

# Expected churn per group under Model 1 (no group structure)
group_data <- group_data %>%
  mutate(
    expected_y_oneparam = n * theta_overall_hat
  )

group_data

plot_compare_models <- group_data %>%
```

```

left_join(theta_summary, by = "Contract") %>%
mutate(
  expected_y_hier = n * mean
) %>%
pivot_longer(cols = c(expected_y_oneparam, expected_y_hier, y),
  names_to = "type",
  values_to = "value") %>%
mutate(type = factor(type,
  levels = c("y",
    "expected_y_hier",
    "expected_y_oneparam"),
  labels = c("Observed",
    "Hierarchical model",
    "One-parameter model"))) %>%

ggplot(aes(x = Contract, y = value, fill = type)) +
geom_col(position = "dodge") +
labs(title = "Observed vs model-implied churn counts by contract",
  x = "Contract type",
  y = "Churn count",
  fill = "Type") +
scale_fill_manual(
  values = c(
    "Observed" = "lightblue4",
    "Hierarchical model" = "lightblue3",
    "One-parameter model" = "lightblue1"
  )
) +
theme_minimal()

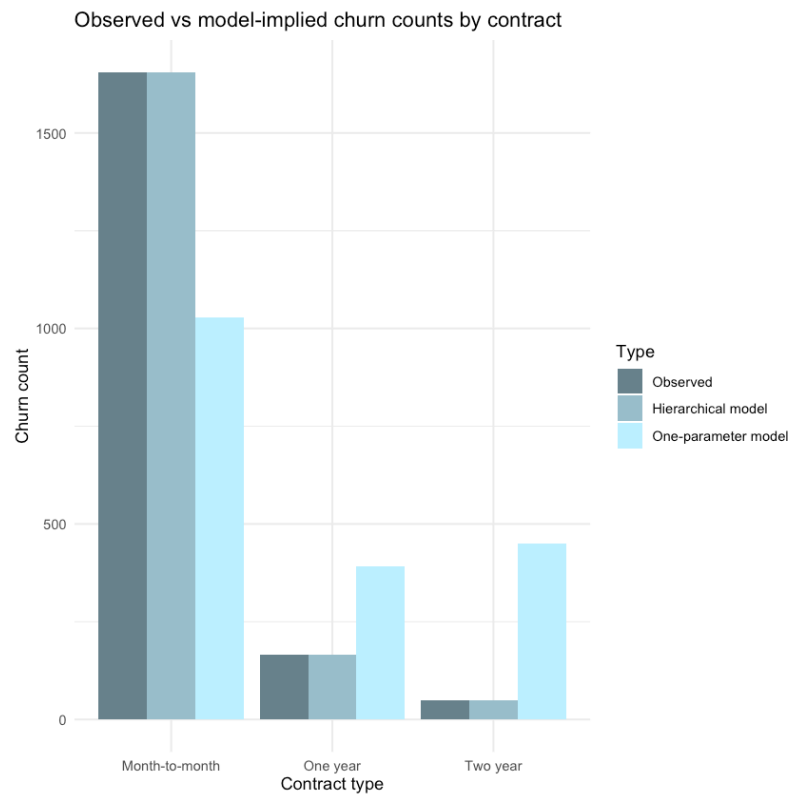
plot_compare_models
ggsave("comparison_models.png", plot_compare_models, dpi = 300)

```

A tibble: 3 × 4

Contract	n	y	expected_y_oneparam
<fct>	<int>	<int>	<dbl>
Month-to-month	3875	1655	1028.5664
One year	1473	166	390.9879
Two year	1695	48	449.9148

Saving 6.67 x 6.67 in image



In []: